

Invisible Invaders Sand Investigation Guide

Team: Piter Garcia and Aastha

Camp: Get Real! Science Camp - Invisible Invaders

Use: Collect, sieve, density-separate, filter, and examine Charlotte Beach sand with campers

Current through: July 22, 2026

TECHNICALLY ACCURATE NAME: We are not filtering sand. We are **sieving the sand, using concentrated saltwater to density-separate material, and filtering the upper saltwater layer** after the heavier sand settles.

EVIDENCE LIMIT: This camp method can reveal and count **suspected fibers, fragments, films, foams, or pellets**. Visual appearance and floating behavior do not chemically confirm that a particle is plastic. Our results are educational screening evidence, not a regulatory measurement or a health test.

Start Here: The Whole Process

LABEL -> COLLECT -> SIEVE -> SORT -> ADD BRINE -> MIX -> SETTLE OVERNIGHT -> FILTER THE UPPER LIQUID -> DRY COVERED -> TAPE-LIFT -> MAGNIFY -> RECORD -> CLAIM + LIMIT

Use this as a three-part investigation rather than rushing an authentic sample:

Camp moment	What campers do	Approximate time
Field Friday	Collect, label, map, and seal three consistent sand samples	10 minutes within the Friday routine
Monday	Sieve, sort visible material, add prepared brine, mix, and begin settling	25-35 minutes
Tuesday	Observe the layers without disturbing them, then filter the upper liquid	15-25 minutes
Wednesday	Examine the dried filters, make tape-lift slides, record suspected particles, compare the blank, and revise the model	30-45 minutes

If the authentic filters will not be dry in time, show campers a clearly labeled **practice or previously prepared filter** for the microscope routine. Do not substitute it silently for their sample.

What Campers Are Learning

1. **A sample represents only a small part of a larger place.** A consistent collection method makes comparisons more meaningful.
2. **Sieves separate materials by size.** Large material stays on larger mesh; smaller material passes to finer mesh.
3. **Salt changes the density of water.** Concentrated saltwater can support some lower-density materials while denser sand settles.
4. **Density separation is incomplete.** Some plastics are denser than saltwater, and weathering or material attached to a particle can change whether it floats.
5. **A filter captures suspended solids; it does not identify them.** Natural material, airborne fibers, and suspected plastics may all remain on the paper.
6. **Scientific claims must match the evidence.** We can report what this sample and method showed while naming what remains uncertain.

Connection To Invisible Invaders

- **EARTH SYSTEMS:** Sand, air, water, living things, weather, and human activity are connected.
- **WATER:** Water can carry material, and the saltwater model lets us investigate movement and separation.
- **LIVING ORGANISMS:** Finding a suspected particle in an environmental sample raises questions about possible encounter pathways; it does not prove exposure or harm.
- **HUMAN INFRASTRUCTURE:** Products, roads, drains, wastewater, waste systems, monitoring, and policy influence where particles come from and where they move.

What Came From Dr. April's Demonstration

The July 9 workshop established the core procedure:

Demonstrated move	Transcript location
Collect the upper approximately 1 cm of sand systematically across a sampling square	July 9 clean transcript, 01:50:33-01:51:34
Stack the sieves largest to medium to smallest; stop before a sieve is more than halfway full; shake consistently	July 9 clean transcript, 01:51:34-01:53:11
Sort visible natural material and suspected human-made material with tweezers	July 9 clean transcript, 01:53:11-01:55:33
Return fine sand to a jar, add saltwater, mix, and let it settle overnight	July 9 clean transcript, 01:55:33-01:57:26
Explain that denser material settles and lower-density material may float, while acknowledging that not every plastic floats	July 9 clean transcript, 01:56:26

Demonstrated move	Transcript location
Put filter paper in a porcelain funnel with tweezers, pour the separated liquid through it, and remember that natural material will also be captured	July 9 clean transcript, 01:58:35-01:59:52
Use clear tape to transfer dried filter evidence to a labeled slide, then examine and measure suspected particles under magnification	July 9 clean transcript, 02:01:48-02:03:52

This guide adds a fixed camp ratio, a procedural blank, written contamination notes, a consistent mixing routine, covered drying, explicit cleanup, and claim-language rules. Those additions make the demonstration easier to repeat and interpret without pretending it is a full analytical laboratory method.

Materials

For three field samples plus one procedural blank

- 4 clean 500 mL glass processing jars with tight lids, labeled **SAMPLE 1**, **SAMPLE 2**, **SAMPLE 3**, and **BLANK**
- the three sealed Field Friday sand jars and map records
- shared **All Camp** lab-quality sodium chloride; this was not an Invisible Invaders purchase
- distilled or deionized water
- at least 1.0 L of adult-prepared concentrated saltwater, plus extra for rinsing or a spill
- graduated cylinder or marked glass measuring cup
- balance, weighing paper or clean metal/glass weighing dish, and scoop for preparing saltwater
- stacked metal sieves, largest mesh on top, then medium, then smallest
- clean metal spoons or scoops, ideally one per sample
- 4 stable trays or clean clipboard backups
- white glass, metal, or clean paper sorting surfaces
- clean tweezers or forceps
- 1-4 porcelain or glass funnels; the current inventory confirms their presence but not the count, so reserve and test them before camp. Ideally use one per sample and blank. If only one is available, process the BLANK first and use the instructor-approved cleaning procedure between samples.
- 90 mm qualitative filter papers verified to fit the funnels
- 4 clean glass receiving jars or beakers
- clean aluminum foil for loose covers
- pre-cleaned microscope slides
- crystal-clear tape and dispensers
- dissecting microscopes or equivalent magnification; one camera/projected view is strongly preferred
- hand lenses and enlarged-image backup
- metric ruler or microscope measurement software

- goggles and powder-free, latex-free gloves in several sizes
- permanent marker, written sample labels, timer, clipboards, and data sheets
- labeled liquid-waste and solid-waste containers approved by the instructors
- visual step cards and role cards

Keep away from the processing area

- glitter, slime, foam beads, yarn, fuzzy craft material, and loose synthetic fabric
- food and drinks
- fans or open windows that blow fibers across uncovered samples
- plastic funnels, plastic sorting trays, and fraying synthetic wipes when glass, porcelain, metal, paper, or foil can do the job
- unrelated open sample jars

Adult Preparation Before Camp

1. Pilot the complete method

Run one practice sample through every step before guiding campers. Confirm:

- the chosen sand amount and brine volume leave at least one-third of the jar empty for mixing;
- lids do not leak;
- the sieve stack, funnels, and filter paper fit;
- filtering does not take longer than the available block;
- filters dry under a loose foil cover;
- clear tape adheres to a dry filter and remains readable under the available microscope; and
- the camera, software, and paper backup all work.

2. Prepare concentrated saltwater

The transcript describes adding salt to water and mixing before adding sand. For a repeatable, no-heat camp preparation:

1. Measure **300 g sodium chloride for each 1.0 L of distilled or deionized water**. A small batch uses **30 g salt per 100 mL water**.
2. Add the salt gradually to water in a clean glass container.
3. Stir or cap and shake until the solution is clear. If a small amount will not dissolve, let it settle and decant only the clear brine.
4. Label the container **LAB SALTWATER - NOT FOR DRINKING**, with the salt mass, water volume, preparer's initials, and date.
5. Keep it covered until use.

Do not call the solution exactly 5 M unless it was prepared to a measured final solution volume. NOAA's full laboratory method uses approximately 5 M sodium chloride with controlled laboratory heating; camp uses a simpler room-temperature concentrated brine and records how it was made.

3. Optional density check

If the balance and graduated cylinder are ready, campers can help verify the brine:

1. Tare a clean container.
2. Measure exactly 50.0 mL of brine.
3. Weigh it in grams.
4. Calculate **density = mass / 50.0 mL**.
5. Record the result instead of assuming every batch is identical.

The teaching point is comparative: concentrated saltwater is denser than fresh water. The class does not need to force the result to a particular number.

4. Establish one camp processing standard

Pilot this starting standard and then use the same values for SAMPLE 1-3:

- **100 mL of the combined bottom two sieve fractions** per processing jar
- **200 mL prepared brine** per jar
- **at least one-third empty headspace**
- **20 slow inversions followed by 30 seconds of controlled shaking**
- **overnight settling**, with start and stop times recorded

If the jar size or available sample requires a different ratio, Piter, Aastha, and the instructors must choose the replacement during the pilot and write it on every data sheet. Do not change the ratio for only one sample.

Safety And Contamination Rules

1. **Nothing is tasted.** Lab salt, saltwater, sand, filters, and particles are not food.
2. **Wear goggles during mixing and pouring.** Use well-fitting gloves or a clean-tool, touch-free route.
3. **Adults check every lid before a camper mixes a jar.** Mix over a tray with two hands.
4. **Adults control broken glass, spills, and disposal.** Campers step back and tell an adult.
5. **Do not heat saltwater with campers.** Do not add hydrogen peroxide, iron catalyst, KOH, bleach, dyes, solvents, or other digestion chemicals to this camp procedure.
6. **Do not use a flame or hot-needle test.** It creates safety, air-quality, and identification problems.
7. **Cover samples whenever the procedure does not require them to be open.** Airborne fibers are evidence contamination, not a camper mistake.
8. **Use one clean tool per sample or clean tools completely between samples.** Never move a spoon directly from SAMPLE 1 to SAMPLE 2.
9. **Record clothing and glove colors.** Similar fibers on the filter and clothing are a contamination question to document.
10. **Contain every suspected particle.** Do not wash captured material back into the environment or an unapproved drain.

Complete Procedure And Teaching Script

Phase 1 - Label Before Opening Anything

Facilitator does

1. Put each sample on its own tray.
2. Label the processing jar, receiving jar, filter-paper envelope or foil cover, slides, and data sheet with the same ID.
3. Prepare **BLANK** exactly like a sample but without sand.
4. Record date, team, sample site, collector, processing people, and clothing/glove colors.

Say: "A label protects the story of the sample. If we lose the label, we lose where the evidence came from."

Ask: "What information would another scientist need in order to repeat what we did?"

Phase 2 - Collect Consistently At The Beach

This phase is already embedded in the [Friday Anchoring Phenomenon Routine](#).

1. Match SAMPLE 1-3 to the map before collecting.
2. Place the sampling square at the instructor-approved site.
3. Use a clean metal scoop to collect the upper approximately **1 cm** in a systematic pattern across the square. Use the same row pattern, depth, and scoop amount at each site.
4. Record site, time, weather, surface condition, nearby activity, tool, and anything that might contaminate the sample.
5. Seal the jar before moving to the next site.
6. Keep SAMPLE 1-3 separate from the prepared MODEL A-D jars.

Say: "We are collecting one small part of the beach. Our sample can answer a question about this jar under this method; it cannot speak for every grain of sand on the beach."

Phase 3 - Sieve By Size

1. Check whether the sand can move freely through the sieves. If it is wet and clumped, air-dry it under a loose clean foil cover. Do not change the method for only one sample.
2. Stack the metal sieves from largest mesh to smallest mesh over the collection pan. Read and record each labeled mesh opening before beginning.
3. Add one measured sample. Stop before the top sieve is halfway full.
4. Close the stack. One camper holds it level while another times **30 seconds** of the same controlled motion used in the pilot.
5. Open one sieve at a time over its labeled tray.
6. Use tweezers to move visible material to **natural/uncertain** or **suspected human-made** areas.
7. Record leaves, twigs, shells, stones, and each suspected fiber, fragment, film, foam, or pellet.
8. Dr. April directed us to proceed with the bottom two sieve fractions and treat them the same. For our three-site camp comparison, combine those bottom two fractions from each site in that site's processing jar and write **COMBINED BOTTOM TWO FRACTIONS** on the jar and data sheet.

If the instructors provide enough jars and time to keep the fractions separate, apply the same complete protocol to both and record the change before processing any sample.

Say: "The sieve tells us about size, not identity. Something small is not automatically plastic, and something natural can be small too."

Ask: "What stayed above? What moved below? What did mesh size cause?"

Phase 4 - Add Brine And Mix

1. Place the pilot-approved amount of sieved sand in its labeled processing jar.
2. Measure the pilot-approved volume of prepared brine and add it to the jar.
3. Leave at least one-third of the jar empty.
4. Seal the jar. An adult checks the lid.
5. Record the sand amount, brine volume, brine batch, and starting time.
6. Over a tray, complete the same mixing routine for every sample: **20 slow inversions, then 30 seconds of controlled shaking.**
7. Process the BLANK beside the samples using the same brine, open-air exposure, tools, timing, funnel, and filter steps, but no sand.

Say: "Salt makes the water denser. Some lower-density material may remain higher while much of the denser sand settles. This is a separation test, not a plastic detector."

Ask before mixing: "Where do you predict the sand and different particles will be after the jar rests? Why?"

Phase 5 - Settle Undisturbed

1. Put each sealed jar upright in its labeled tray on a stable shelf away from traffic and direct heat.
2. Record the settling start time.
3. Do not move, tap, open, or compare jars by picking them up.
4. Let the jars settle overnight.
5. The next day, observe before touching. Photograph or draw the surface, water column, and bottom layer.
6. Record the end time and total settling time.

Say: "We are giving gravity time to separate material. A particle that sinks is not proven natural, and a particle that floats is not proven plastic."

Ask: "What changed while nobody touched the jar? What evidence supports your explanation?"

Phase 6 - Filter The Upper Saltwater Layer

Process BLANK first, then SAMPLE 1-3.

1. Set a clean porcelain or glass funnel over the matching glass receiving jar.
2. Use clean tweezers to place the matching filter paper into the funnel. Avoid touching the filter surface.
3. Without shaking or remixing, bring the processing jar to the tray.

4. Carefully pour the **upper saltwater layer** through the filter. Stop before settled sand enters the funnel.
5. If floating material sticks to the upper jar wall, use a small, recorded rinse of distilled water to move it onto the same filter.
6. Let all liquid pass through. Do not press, scrape, or rub the filter.
7. Use tweezers to transfer the filter to its labeled clean surface.
8. Cover loosely with clean aluminum foil and air-dry for 12-24 hours.
9. Keep the filtrate and settled sand contained until an instructor approves disposal.

Say: "The filter catches suspended material. It can catch natural material, airborne fibers, and suspected plastics. The filter does not tell us which is which."

Ask: "What passed through? What stayed on the paper? What can the filter answer, and what can it not answer?"

Phase 7 - Make A Tape-Lift Slide

Only use a completely dry filter.

1. Label the slide with sample ID, date, filter fraction, and team.
2. Divide the observation plan into four filter areas: top-left, top-right, bottom-left, and bottom-right. Do not write across the evidence surface.
3. Press one clean piece of crystal-clear tape lightly onto the first area; do not rub.
4. Lift once and place the tape flat on the labeled slide.
5. Repeat with fresh tape or a new slide for each planned area.
6. Record which filter area each tape lift represents.

Say: "A systematic path helps us look beyond the first interesting thing we notice."

Phase 8 - Magnify, Classify, And Count

1. Begin with the BLANK slide, then examine SAMPLE 1-3.
2. Use approximately 40X magnification when available, or record the actual magnification used.
3. Scan in a consistent row-by-row path rather than jumping around the slide.
4. For every suspected particle, record:
 - o form: fiber, fragment, film, foam, or pellet;
 - o color;
 - o longest visible length in millimeters when measurement is possible;
 - o slide area or tape-lift number;
 - o confidence: **likely natural**, **uncertain**, or **suspected human-made**; and
 - o one reason for the classification.
5. A second camper or facilitator checks at least one slide area. Record disagreements rather than erasing them.
6. Use the microscope camera or shared display so nobody must crowd an eyepiece.

Say: "We are describing what the particle looks like. Without chemical analysis, our strongest honest label is suspected plastic."

Ask: "What observation supports your classification? What else could this particle be?"

Phase 9 - Compare The Blank Before Making A Claim

1. Count and describe material found on the BLANK filter.
2. Compare blank colors and forms with SAMPLE 1-3.
3. If a similar fiber appears in the blank and samples, flag it as a possible handling or airborne contaminant.
4. Do not automatically subtract the blank count. Report the blank beside the sample and explain how it affects confidence.

Say: "The blank helps us notice what our tools, air, clothing, or handling may have added. It does not accuse anyone; it helps us improve the method."

Phase 10 - Make A Claim That Fits The Evidence

Use this frame:

In SAMPLE ____, processed with ____ mL sand and ____ mL brine after ____ hours of settling, we observed ____ suspected fibers/fragments/films/foams/pellets. The BLANK contained _____. This evidence suggests _____. We cannot yet conclude ____ because ____.

Acceptable camp claim:

"Under our camp method, SAMPLE 2's filter contained three suspected blue fibers. The blank contained one similar blue fiber, so some of the evidence may have entered during handling or from the air. Visual observation cannot confirm the material."

Claims the method does **not** support:

- "There are exactly three microplastics at Charlotte Beach."
- "Every floating particle was plastic."
- "No floating particles means there was no plastic."
- "This particle came from a particular person, neighborhood, product, or company."
- "This sample proves that someone was exposed or harmed."

Camper Roles And Access Routes

No role owns the science. Campers may switch, share, pass, or return.

Role	Scientific contribution	Access route
Label protector	Matches every jar, filter, slide, and data sheet	Seated, low-movement
Sampler	Uses the consistent field pattern	Movement or partnered
Sieve operator	Runs the same timed motion	Standing, seated-table, or partner-assisted

Role	Scientific contribution	Access route
Timer	Protects mixing, settling, and filtering time	Visual, audio, or tactile timer
Brine measurer	Reads or verifies volume	Large-print scale or partner reading
Filter monitor	Watches flow and tells the pourer when to stop	Touch-free observation
Evidence photographer	Records layers, labels, and non-identifying evidence	Camera or tablet
Microscope navigator	Moves through the slide systematically	Eyepiece, shared screen, or enlarged still image
Evidence recorder	Draws, writes, types, dictates, points, or records audio	Multiple communication modes
Skeptic/checker	Asks what else a particle could be and checks the blank	Verbal or nonverbal
Seated coordinator	Tracks the whole sequence and next step	Low-energy route

Facilitators show the full visual sequence while giving **one spoken direction at a time**. Offer goggles that fit, several glove sizes, a no-touch route, seated work, quiet observation, a shared microscope display, breaks, home language, partner response, and optional public speaking.

Data Record

Use one row per sample or fraction.

Field	Record
Sample ID and map site	
Date, collection time, and weather	
Collection depth, pattern, and scoop count	
Wet, damp, or dry when sieved	
Sieve mesh openings and COMBINED BOTTOM TWO FRACTIONS	
Sand amount processed	
Salt mass and water volume in brine batch	

Field	Record
Measured brine density, if checked	
Brine volume added	
Mixing method and duration	
Settling start, stop, and total time	
Filter ID and filter-paper type	
Rinse volume, if used	
Actual magnification	
Suspected forms, colors, lengths, and counts	
Blank observations	
Possible contamination events	
Claim	
Evidence limit	
Model change or new question	

Troubleshooting

Problem	Do this	Do not do this
Wet sand clogs the sieve	Cover and air-dry all comparable samples; record the change	Force only one sample through or add unrecorded water
Salt remains at the bottom of the brine container	Let it settle and use the clear upper brine; record the recipe	Pour salt crystals into one sample only
Processing jar leaks	Stop, contain the spill, transfer only with adult approval, and flag the sample	Keep shaking or hide the loss
Sand begins entering the funnel	Stop pouring; record it; use a new filter if flow is blocked	Scrape or squeeze the filter

Problem	Do this	Do not do this
Filter flows very slowly	Let gravity work; cover the setup; use the same filter type for all samples	Poke holes or press material through
Filter is still wet	Keep it covered and wait, or use a clearly labeled practice slide	Apply tape and pretend the poor transfer is sample evidence
Fibers appear in the blank	Report them and lower confidence in similar sample fibers	Delete the blank or automatically subtract counts
Microscope cannot connect	Use a hand lens, enlarged still image, drawing, and preserved filter	Treat technology failure as learner failure
Campers disagree about a particle	Record both classifications and the evidence for each	Let the fastest or loudest answer become fact

Cleanup And Disposal

1. Keep every filter, tape lift, and suspected particle labeled and covered.
2. Place used filters and tape in a labeled sealed solid-waste container or archive envelope.
3. Do not pour unfiltered sample liquid, sand, or captured solids down a drain.
4. Follow Dr. April's or the lab's approved disposal route for filtered brine, environmental sand, and glassware wash water.
5. Wash reusable glass, porcelain, and metal tools with the approved laboratory method; rinse with distilled water; air-dry covered.
6. Clean one sample station at a time to prevent label swaps.
7. Photograph completed data sheets and preserve the original sheets.
8. Wash hands after gloves are removed and before eating.

Facilitator End Check

- ☐ Every jar, filter, slide, and data sheet has a matching written ID.
- ☐ SAMPLE 1-3 and BLANK used the same documented procedure.
- ☐ Sample amount, brine amount, mixing time, and settling time are recorded.
- ☐ Samples stayed covered except during required handling.
- ☐ The blank was examined before claims were written.
- ☐ Campers used **suspected**, **observed**, and **under our method** language.
- ☐ Nobody claimed visual confirmation, exact beach prevalence, a source, exposure, or harm.
- ☐ At least one seated, touch-free, quiet, and shared-screen route remained available.
- ☐ All sample material and liquid followed the approved containment and disposal plan.
- ☐ Campers added one evidence-supported relationship, limitation, or question to the Invisible Invaders model.

Camper Quick Guide

Day 1: Separate

1. **MATCH** the sample ID on the jar, map, and data sheet.
2. **MEASURE** the same amount of the combined bottom two sieve fractions used for the other samples.
3. **SIEVE** for the same amount of time.
4. **SORT** what you see as natural/uncertain or suspected human-made.
5. **ADD** the same amount of prepared saltwater.
6. **MIX** with the class routine over a tray.
7. **PREDICT** what will rise, stay in the middle, or sink.
8. **SETTLE** overnight without touching the jar.

Day 2: Filter

1. **NOTICE** the layers before moving the jar.
2. **SET** clean filter paper in the matching funnel.
3. **POUR** only the upper saltwater layer; leave settled sand behind.
4. **COVER** the filter and let it dry.

Day 3: Examine

1. **LABEL** the slide.
2. **LIFT** evidence from one planned filter area with clear tape.
3. **SCAN** the slide in rows.
4. **DESCRIBE** form, color, size, and confidence.
5. **COMPARE** with the BLANK.
6. **CLAIM** only what the evidence supports.
7. **NAME** what is still unknown.
8. **REVISE** the class model or add a new question.

Scientific Foundation

- [NOAA: Laboratory Methods for the Analysis of Microplastics in the Marine Environment](#) supports density separation, overnight settling, transfer of floating solids, covered drying, and microscope examination. NOAA's complete analytical method includes heated and reactive laboratory steps that are **not part of this youth camp protocol**.
- [EPA: National Coastal Condition Assessment Field Operations Manual](#) supports glass jars, metal sampling tools, recording clothing and glove colors, minimizing synthetic-fiber exposure, and collecting an air blank when sediment is exposed.
- [EPA: Handbook for Citizen Science Quality Assurance and Documentation](#) supports consistent written procedures, field and equipment blanks, replicate or split samples, training, and matching the strength of claims to the intended use and quality of the data.

AI Use Disclosure

AI was used under Piter Garcia's supervision and guidance to make this guide more inclusive and accessible by organizing, structuring, formatting, and enhancing content Piter provided. Piter reviewed and guided all content; AI helped validate direct sources and identify missing safety, contamination-control, and evidence-limit steps. Piter remains responsible with Aastha and the instructors for piloting and camp use.